

1.0 Introduction

This manual covers routine maintenance, safety lockout procedures, and compliance requirements for the Cooling Water System, including Gate Valve V-104 and Centrifugal Pump P-201A.

1.1 Scope

Applies to all plant personnel performing maintenance, inspection, or isolation work on cooling loop equipment tagged V-104 and P-201A.

1.2 Referenced Standards

OSHA 1910.147 - The Control of Hazardous Energy (Lockout/Tagout).

OSHA 1910.119 - Process Safety Management of Highly Hazardous Chemicals.

2.0 Equipment Overview · Valve V-104

V-104 is a manually operated gate valve isolating the cooling water supply line upstream of Pump P-201A. It is the primary isolation point for downstream pipe maintenance.

2.1 V-104 Lockout/Tagout Procedure

Prior to any pipe maintenance on the cooling loop, valve V-104 shall be closed, locked out, and tagged in accordance with the plant LOTO program. Verify zero energy state before proceeding.

2.2 V-104 Reassembly Torque Specification

When reassembling V-104, torque the bonnet bolts to 45 Nm following a star/cross pattern in two passes to ensure even seating of the gasket.

2.3 V-104 Regulatory Compliance

All isolation of V-104 must comply with OSHA Standard 1910.147
(The Control of Hazardous Energy).

3.0 Equipment Overview · Pump P-201A

P-201A is a centrifugal pump circulating cooling water through the condenser loop. It draws suction through V-104.

3.1 P-201A Lubrication Schedule

Pump P-201A bearings require re-lubrication with ISO VG 68 oil every 400 operating hours. Do not over-fill; maintain the oil sight glass at the center mark.

3.2 P-201A Dry-Run Prevention

Before starting P-201A, confirm suction valve V-104 is fully open.
Running the pump dry will destroy the mechanical seal within seconds.

3.3 P-201A Regulatory Compliance

P-201A is covered equipment under the facility's OSHA 1910.119
Process Safety Management mechanical integrity program.

4.0 Inspection Checklist

Quarterly: visual inspection of V-104 packing gland for leaks.

Quarterly: verify P-201A bearing temperature within spec.

4.1 Spare Parts

V-104: gasket kit GK-104, packing set PK-104.

P-201A: mechanical seal MS-201, bearing kit BK-201.

5.0 Emergency Shutdown

In the event of abnormal vibration or leakage from P-201A, close V-104 immediately and notify the shift supervisor.

6.0 Revision History

Rev A - Initial release for ET AI Hackathon 2026 Golden Dataset.